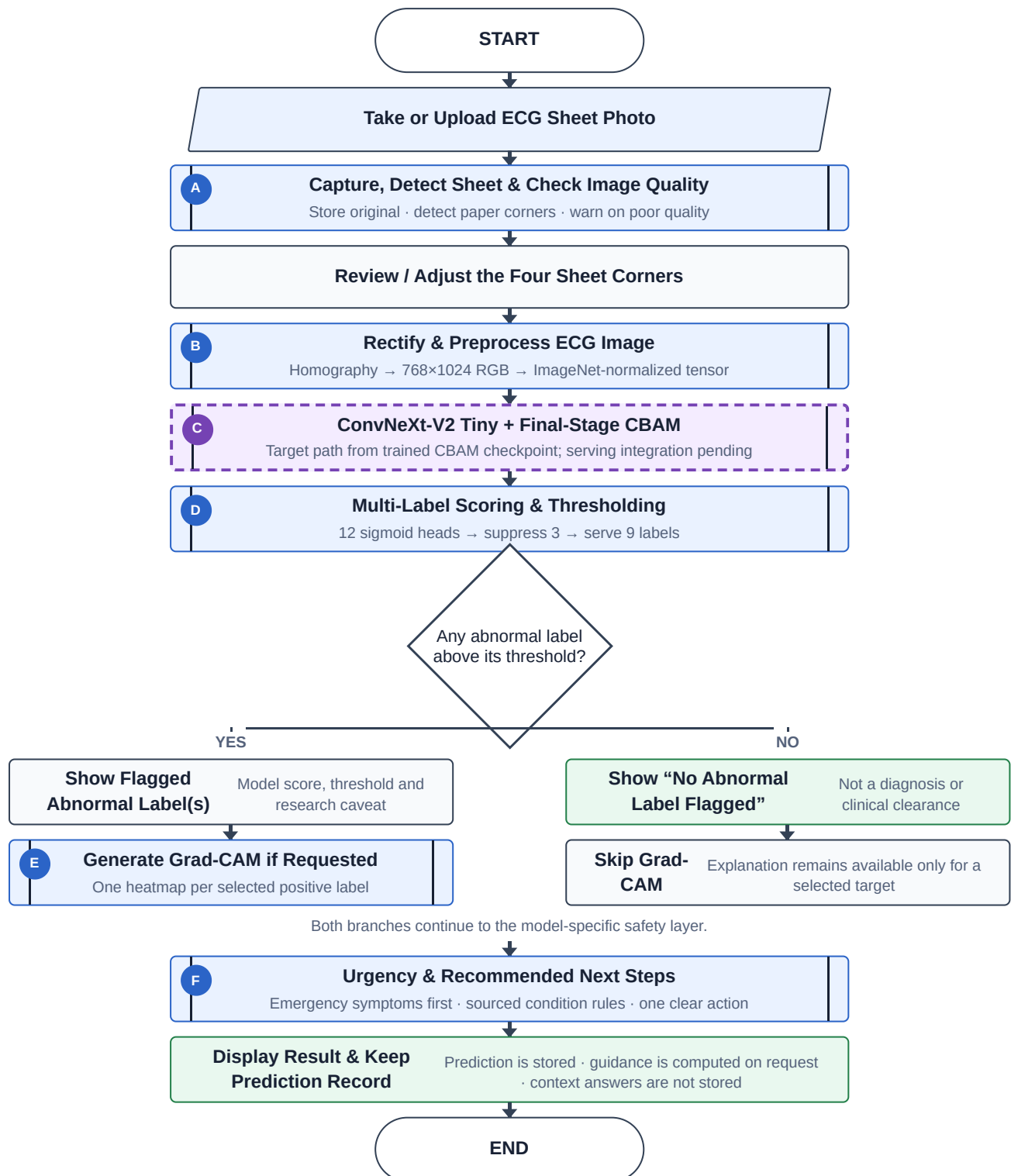


Main CardioSentry System Flowchart

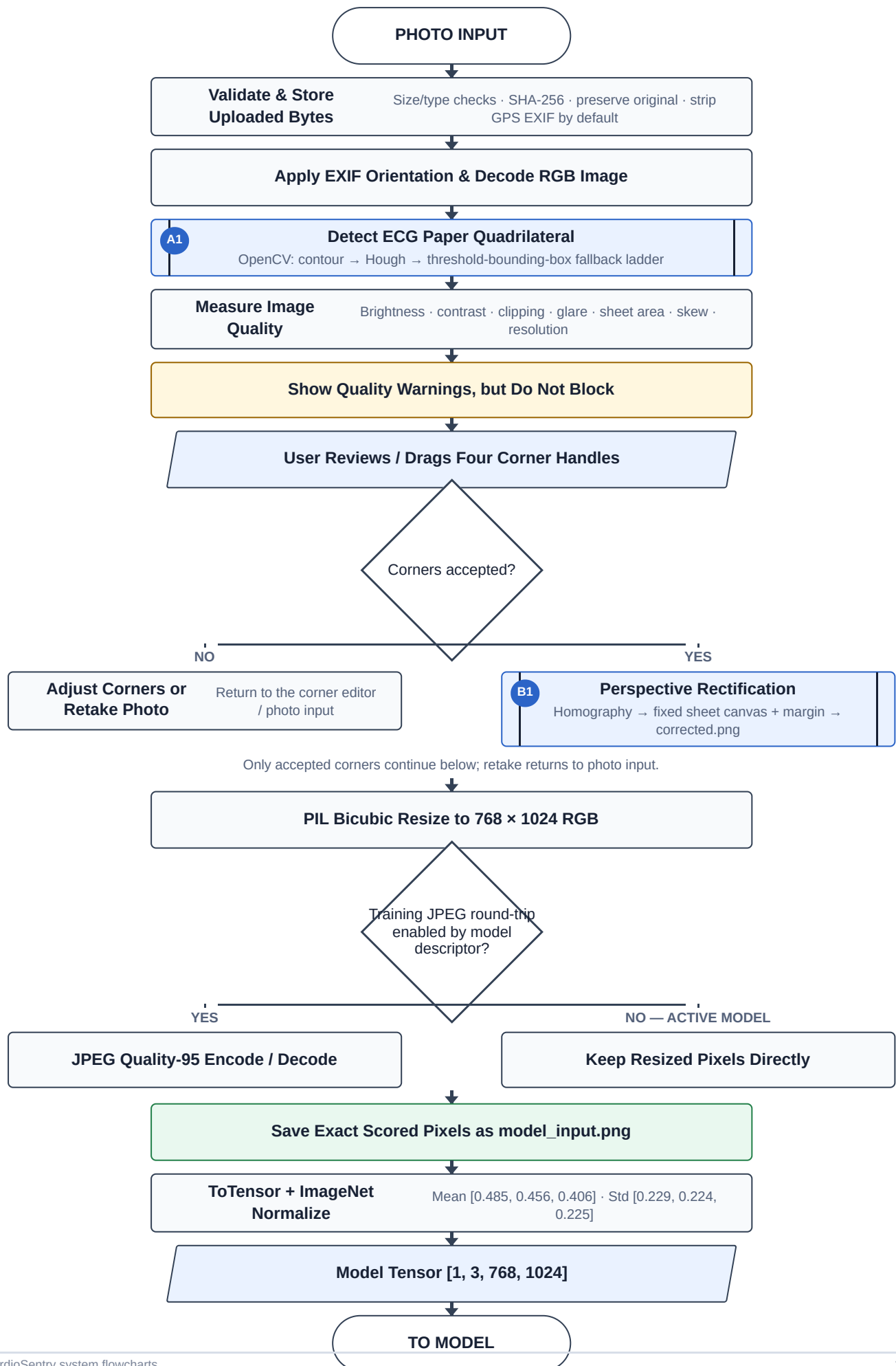
Direct application flow, with the pending CBAM model path and implemented guidance stage clearly marked.

Implemented in the application Trained artifact exists; app integration pending



A–B. Capture, Detection, Rectification & Preprocessing

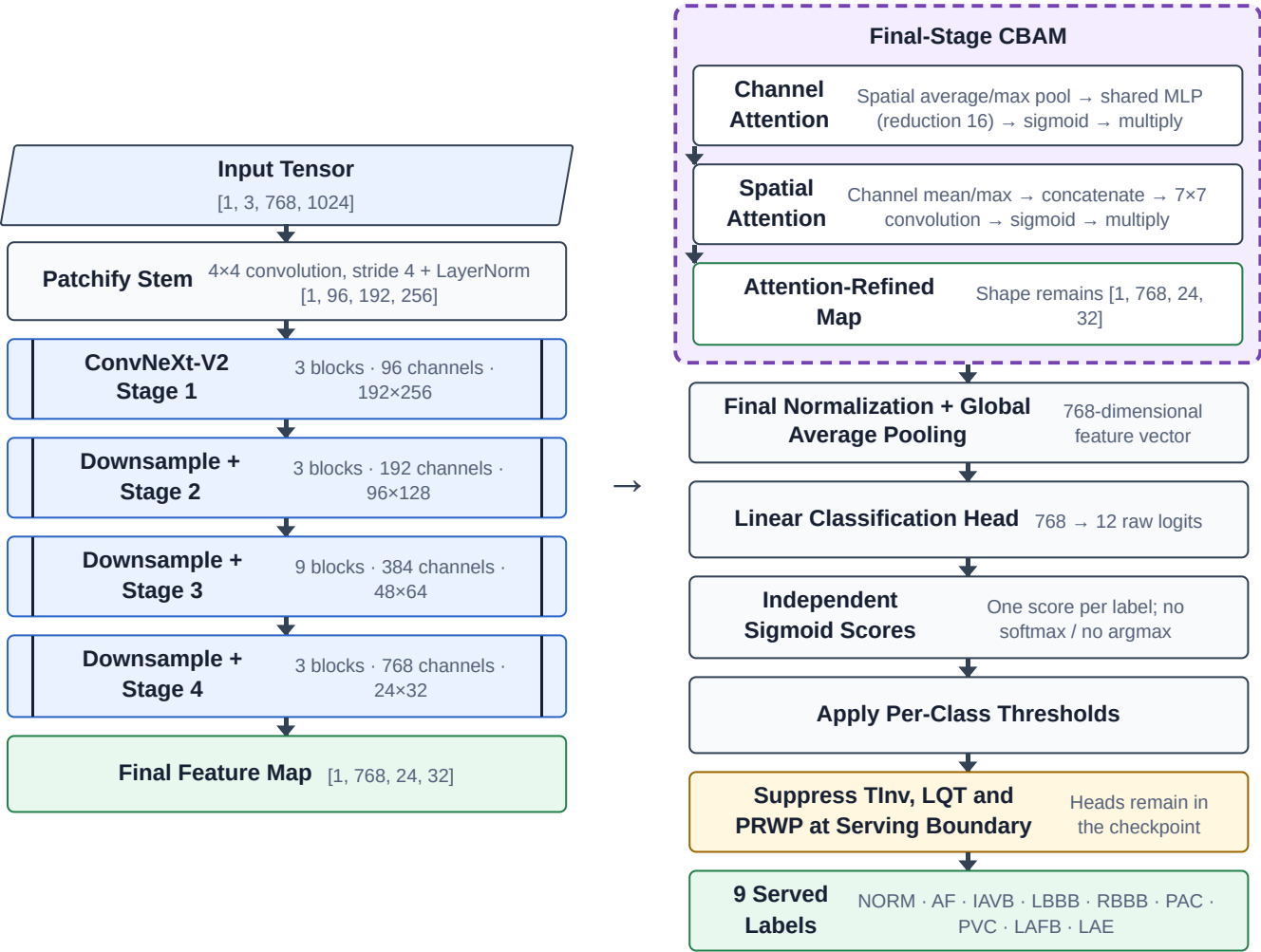
What happens to the photograph before it reaches the model.



C-D. ConvNeXt-V2 Tiny + CBAM Classification

Requested model path: hierarchical feature extraction, attention refinement and nine served outputs.

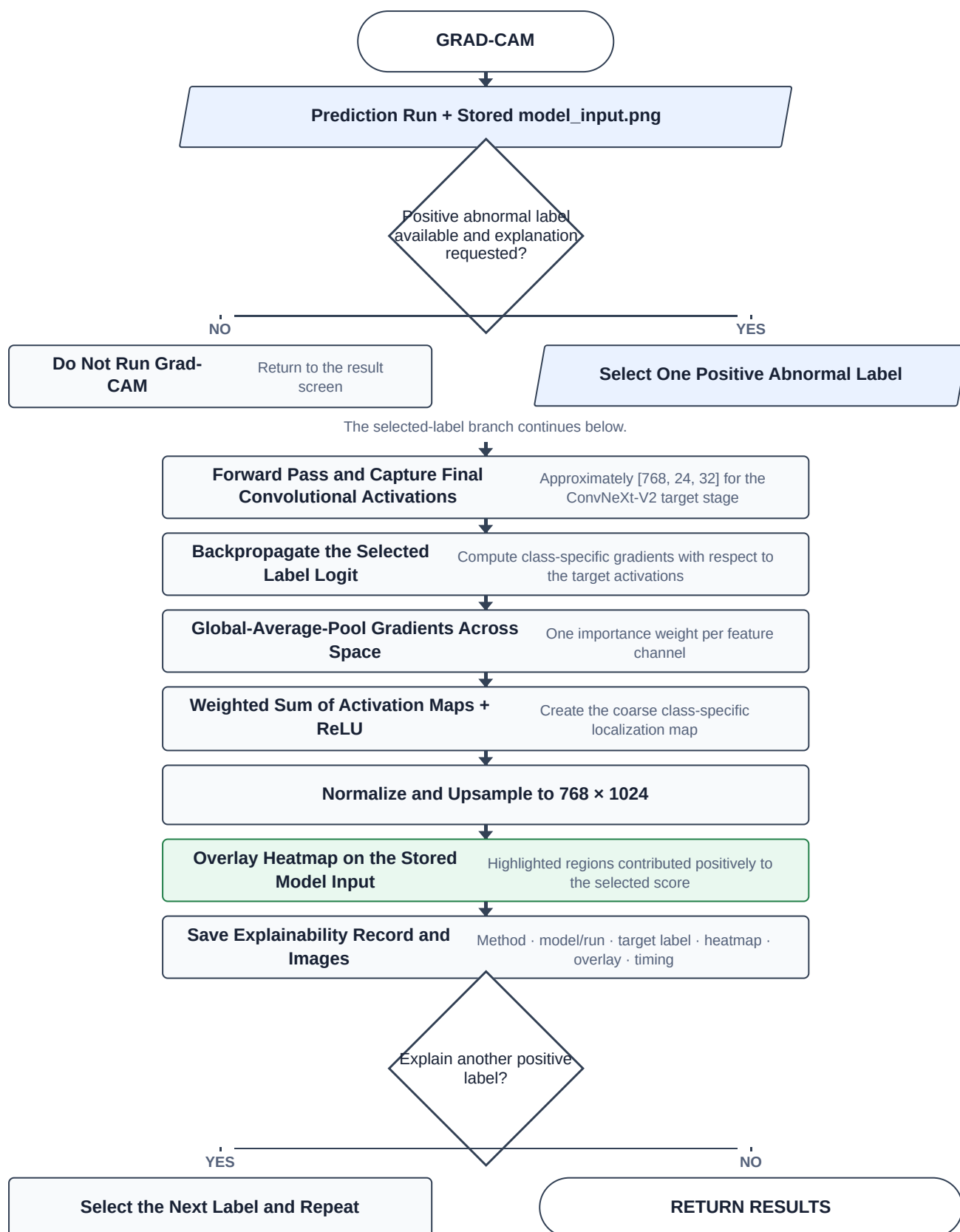
Implementation status: this CBAM architecture is trained and checkpointed in the research tree. The running application still serves the bare ConvNeXt-V2 Tiny descriptor; the CBAM wrapper, descriptor and active-model switch remain to be integrated.



Output wording: show “model score” and “flagged label,” not a confirmed diagnosis or calibrated probability. Real-photo calibration remains unestablished.

E. Grad-CAM Explainability Flowchart

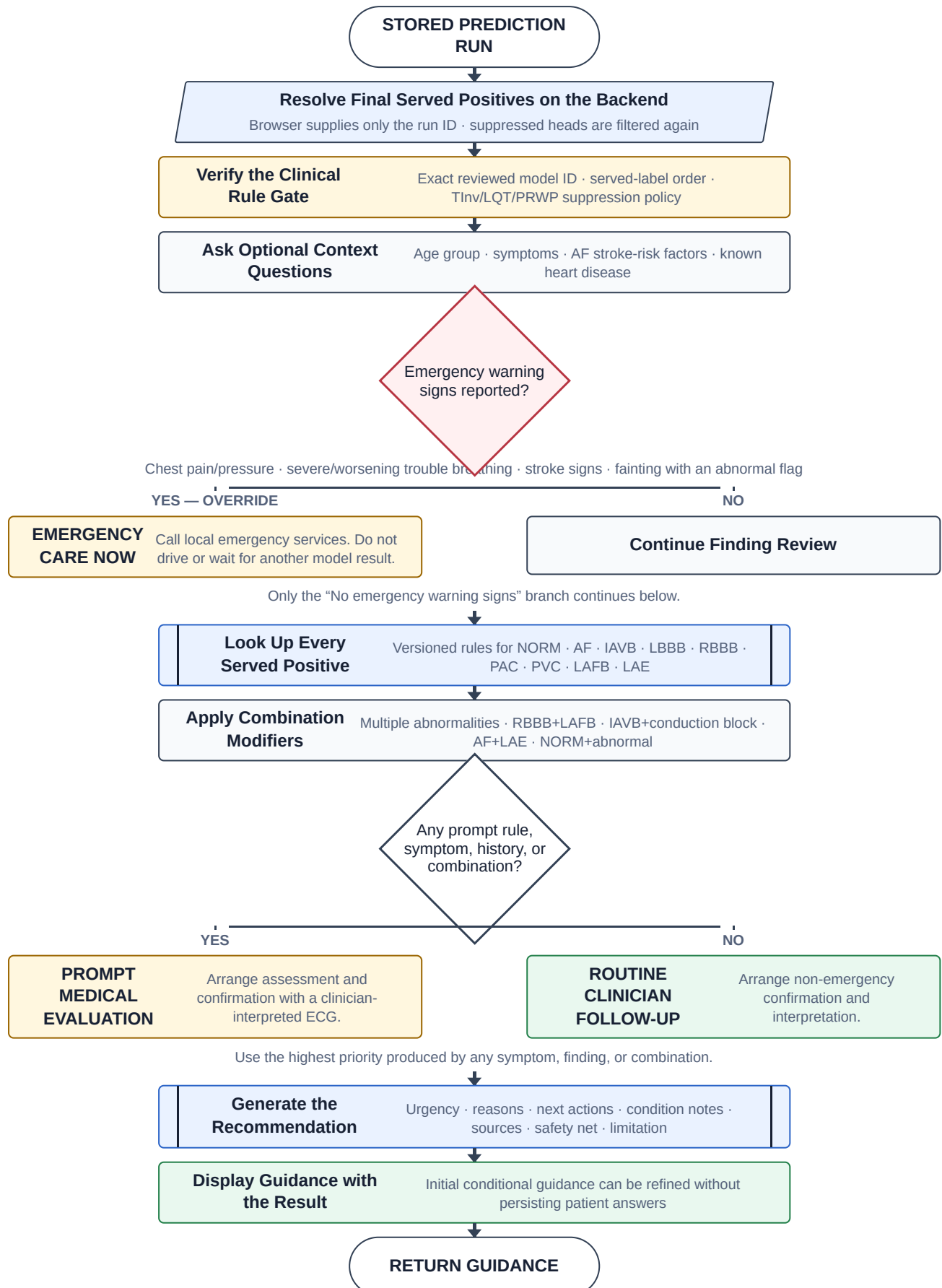
Post-hoc explanation for a user-selected positive abnormal label; it never changes the prediction.



Interpretation: Grad-CAM shows influential image regions, not whether the model is correct and not which waveform feature a clinician would diagnose.

F. Urgency & Recommendation Layer

Implemented for the current model's nine served labels; convert stored positives and optional patient context into sourced next-step guidance.



Required safeguards: a higher model score does not mean greater clinical urgency. Symptoms can override the label rules; findings come only from the stored run; suppressed heads never receive guidance; and the output is screening support, not diagnosis or treatment.